

Effects of Understory Vegetation Management on Brood Habitat for Northern Bobwhites

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ABSTRACT Northern bobwhite (*Colinus virginianus*) populations have experienced severe declines for several decades, and declines have been particularly precipitous in the southern United States. These declines are partially attributable to large-scale conversions of potential habitat to short-rotation pine (*Pinus* spp.) forests managed for wood fiber production and fire exclusion in pine-dominated landscapes. We used standard arthropod sampling techniques, human-imprinted bobwhite chicks, and vegetation response to evaluate effects of different understory vegetation management practices on brood habitat quality within a commercially managed pine forest in Louisiana, USA, during 2002–2005. Specifically, we evaluated effects of mowing, prescribed fire during the growing season, prescribed fire in combination with imazapyr application, and no vegetation management on arthropod abundance and diversity, vegetation response, and the probability of bobwhite chicks successfully capturing an arthropod. Bobwhite chicks were more successful at capturing arthropods, and arthropod abundance and diversity were greatest, in plant communities managed using prescribed fire and imazapyr. Forest stands managed using a combination of fire and imazapyr were managed primarily to benefit the federally endangered red-cockaded woodpecker (*Picoides borealis*, RCW). Our findings suggest that management directed toward improving forest condition for RCWs improves habitat quality for brooding bobwhites. However, bobwhite chicks in our study area were less successful at capturing arthropods than were chicks in other studies in the southeastern United States. Brood-rearing habitat in pine forests similar to those we studied may be of generally poor quality, and could be related to precipitous declines of bobwhites in the West Gulf Coastal Plain. Managers should recognize that creating high-quality brood habitat in forests similar to those we studied will require more intensive management of understory vegetation than we studied. (JOURNAL OF WILDLIFE MANAGEMENT 72(6):1361–1368; 2008)

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Northern bobwhite (*Colinus virginianus*; hereafter bobwhite) populations have experienced drastic declines since the early 1960s (Droege and Sauer 1990), with the most precipitous declines occurring in the southern United States (Rosene 1969, Brennan 1991, Church et al. 1993). For instance, in Louisiana, USA, bobwhite populations declined 4.9% annually from 1982 to 2003 (Sauer et al. 2004). Bobwhite populations in Louisiana are primarily part of the West Gulf Coastal Plain region encompassing land in 4 southeastern states (AR, TX, OK, LA), which has suffered one of the most significant annual decline trends (–8.2%) of all the bird conservation regions in the United States (Dimmick et al. 2002). This decline is attributed primarily to deterioration and loss of suitable herbaceous ground cover associated with large-scale changes in agriculture and forestry, as well as fire exclusion from pine (*Pinus* spp.)-dominated forests (Roseberry and Klimstra 1984, Brennan 1991).

Historically, southern pine forests of the lower Gulf Coastal Plain were dominated by longleaf pine (*Pinus palustris*) overstories with an herbaceous understory maintained with regular fire during the growing season (Frost 1993). More than 12.9 million ha of southern forests have been converted to pine plantations (Wear and Greis 2002), which contain pine species that grow faster than longleaf, have shorter stand rotations (20–30 yr), are planted at higher

densities, and lack disturbance regimes (e.g., regular fire rotations) that retain herbaceous ground cover (Brennan 1991). Without the use of fire or other disturbance to maintain herbaceous ground cover, landscape quality for bobwhites will decline rapidly through natural succession (Wilson et al. 1995, Burger 2002).

The red-cockaded woodpecker (*Picoides borealis*; RCW), a pyric-adapted species, is listed as a federally endangered species and has been the impetus for changes in fire regimes across expanses of southern pine forests (United States Forest Service 1995). Intensive management of RCW colonies includes the maintenance of low-basal-area pine forests using short-rotation (1–3 yr) mowing and burning regimes, often in conjunction with herbicide treatments, to reduce encroachment of woody species (Bowman et al. 1999). Management directed at improving conditions for bobwhites in southern pine forests parallels management for RCWs (Brennan 1991, Wilson et al. 1995), and management strategies implemented for RCWs may positively affect foraging conditions for bobwhites through increases in arthropod biomass (Hurst 1972, Fuller 1994) and bobwhite abundance at multiple spatial scales (Cram et al. 2002, Chamberlain and Burger 2005).

During 2002–2005, we conducted a field study to evaluate how vegetation management practices commonly used across commercially managed pine forests affect brooding habitat for bobwhites. Our objective was to evaluate effects of understory mowing, prescribed fire during the growing

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season, imazapyr application in conjunction with fire during the growing season (a consequence of management focused on RCWs), and no treatment on understory vegetation within managed pine forests.

STUDY AREA

We conducted research on the 13,137-ha Jackson-Bienville Wildlife Management Area (JBWMA) in Jackson, Bienville, and Lincoln parishes of north-central Louisiana. The JBWMA was bordered on the east by United States Highway 167 and on the west by Louisiana Highway 147. The JBWMA was owned primarily by Weyerhaeuser Company, with the principal land use being production of timber and wood fiber. The area was traversed with >65 km of gas pipelines and rights-of-way maintained by the Louisiana Department of Wildlife and Fisheries as food plots planted annually for wildlife. Mean annual rainfall was 123.7 cm and mean annual temperature was 18.2° C.

Forests on JBWMA were composed primarily of even-aged, short-rotation (25–40 yr) loblolly pine (*Pinus taeda*) plantations. Approximately 10–20% of JBWMA was considered to be bottomland hardwood forests, containing Streamside Management Zones and seasonally flooded low areas adjacent to rivers and streams. Topography on JBWMA was gently rolling hills, consistent with an upland coastal plain site. Primary overstory species throughout forested stands included loblolly and shortleaf pine (*Pinus echinata*), red maple (*Acer rubrum*), water oak (*Quercus nigra*), southern red oak (*Q. falcata*), mockernut hickory (*Carya tomentosa*), and sweetgum (*Liquidambar styraciflua*). Understory plant communities were dominated by a variety of herbaceous species, as well as blackberry (*Rubus* spp.), greenbrier (*Smilax* spp.), poison ivy (*Toxicodendron radicans*), French mulberry (*Callicarpa americana*), yaupon (*Ilex vomitoria*), sumacs (*Rhus* spp.), lespedezas (*Lespedeza* spp.), and beggarweeds (*Desmodium* spp.).

METHODS

Stand Treatments

We selected loblolly pine stands >20 years old (range = 24–46 yr) to evaluate effects of different understory vegetation management practices on bobwhite brood habitat. We selected stands based on the criteria that they had received no prior understory management, had been commercially thinned ≥ 3 years prior, and contained advanced hardwood or woody succession in the understory. Commercial thinning was designed to reduce tree density to approximately 300–325 trees/ha, although intensity of thinning often varied across and within individual forest stands. Vegetation management practices included mowing, prescribed fire only, imazapyr application in conjunction with prescribed fire, and no treatment. We chose these practices because they were practiced by private and public land-owners on and adjacent to our study area. Stands that received mowing treatments were bush-hogged in April–May of 2003 and 2005 and are hereafter referred to as mowed stands. Mowed stands were mowed to an approx-

imate vegetation height of 10 cm during each treatment, then were allowed to succeed naturally. Stands that received prescribed fire only were burned during the growing season (Apr–May) in 2003 and 2005 to simulate a 2-year burn rotation and are hereafter referred to as burned stands.

As part of efforts to improve habitat for RCWs on the study area (active colonies were present at the beginning of the study, although not in the stands included in our study), selected stands were slated to have herbicide applications followed by prescribed fire to reduce hardwood midstory vegetation and increase herbaceous vegetation in the understory (i.e., to produce foraging habitat for RCW). These stands received applications of imazapyr (Arsenal AC®, Cary, NC) at a rate of 226 g of imazapyr/0.405 ha in September or October for optimum timing of herbicide application (Miller 1989). We applied imazapyr along with water at 185 L/ha plus a 0.5% nonionic surfactant. We broadcast sprayed herbicides with a T-boom system mounted on a skidder. Once stands were treated with imazapyr, they were subsequently burned during the following growing season (Apr–May) to remove standing and downed debris and stimulate herbaceous plant growth. These stands are hereafter referred to as burn–herbicide stands. Burn–herbicide stands were treated during 2001 (prior to our study) and 2002 and burned on a 2-year rotation. Stands that received no manipulation to the understory since 2000 are hereafter referred to as control. We applied each vegetation management practice to 6 forest stands, ranging from 2 ha to 13 ha in area and repeatedly sampled these stands each year. Hence our study included 24 forest stands (4 vegetation management practices replicated in 6 forest stands) distributed throughout the study area.

Bobwhite Chick Foraging Trials

Palmer et al. (2001) estimated the consumption rate of arthropods by imprinted bobwhite chicks foraging in a variety of habitat patches, and Smith and Burger (2005) reported that using commercially available, imprinted bobwhite chicks was a reasonable method for indexing arthropod availability and brood habitat quality for wild bobwhite chicks. We used the general methodology outlined by Palmer et al. (2001) and Smith and Burger (2005) for imprinting chicks and conducting foraging trials.

We obtained pen-strain bobwhite eggs from a local commercial producer (Moon's Quail Farm, Homer, LA) who collected eggs from birds held on-site. We used a standard incubator (Model 1502; Georgia Quail Farms, Savannah, GA) to incubate 100–150 eggs 3 times during each year (Jun, Jul, Aug; hereafter brooding periods) at 37.5° C for 21–24 days. We immediately transferred posthatch chicks to a brooder unit (Model 0701; Georgia Quail Farms), which contained heat lamps that maintained a constant temperature of 37.5° C.

We considered chicks successfully imprinted when they readily responded to our whistle calls and followed us. Twice daily we took out successfully imprinted chicks that were 2–11 days old for 1-hour training periods to expose them to a

variety of habitats similar to the stands used in our study and to familiarize them with arthropod foraging. Between these training periods, we maintained chicks in brooders and fed them a commercial diet of poultry starter feed.

We performed foraging trials within each stand receiving understory vegetation management during June–August annually, which encompassed most of the brooding season for bobwhites in north-central Louisiana. We separated trials into the 3 brooding periods (Jun = early, Jul = middle, Aug = late) to evaluate temporal differences in chick foraging efficiency relative to vegetation management techniques. We conducted trials with successfully imprinted chicks that were 11–14 days old. We restricted arthropods from the chicks' diets for 18 hours prior to foraging trials and we restricted all food for 4 hours before trials began to clear all crops and gizzards of arthropod parts and encourage foraging during trials. We released 2 broods (5 chicks/brood when possible) into an approximately 1-ha area (hereafter plot) within each stand between 0900 hours and 1600 hours on dry days only to standardize sampling. Within each plot, we placed one brood near the center and the other near the edge. Handlers remained stationary and did not allow chicks to forage behind them to avoid disturbing arthropods on the foliage. We allowed chicks to forage for 30 minutes, then euthanized them using CO₂ asphyxiation or cervical dislocation and immediately froze them for crop analysis in the lab. We rinsed contents of the crop and esophagus and emptied them into 120-ml sample cups containing ethyl alcohol and identified items to taxonomic order with a 30× microscope. We conducted research under Louisiana State University Agricultural Center Institutional Animal Care and Use Committee (Protocols AE 02-07 and AE 03-04).

Arthropod Abundance and Diversity Sampling

We used sweep net and pitfall trap samples to assess relative abundance and diversity of arthropods, whereas foraging trials served as a measure of availability of arthropods to chicks relative to each vegetation management practice. We then compared chick crop estimates to abundance and diversity estimates (i.e., sweep net, pitfall) to evaluate differences among vegetation management practices relative to brood habitat quality.

We determined arthropod abundance using a 38-cm-diameter sweep net and pitfall traps within the same plots used for the chick foraging trials. We collected sweep net samples immediately following chick foraging trials along 3 nonoverlapping 5-m transects (approx. 10–15 sweeps). We immediately froze arthropods collected with sweep nets, stored them in 120-mL sample cups containing ethyl alcohol, and later identified them to taxonomic order. We grouped all arthropod orders that occurred infrequently together and included them in the category “Other” (i.e., Dermaptera, Dictyoptera, Ephemeroptera, Homoptera, Mantodea, Phasmatodea, and Thysanura).

We constructed 3 pitfall traps along a diagonal transect through the middle of each plot. We set traps immediately following chick foraging trials. Each pitfall was constructed of one 400-mL plastic beaker, which held a smaller,

trimmed-to-fit 250-mL beaker containing ethylene glycol. We placed traps in the ground prior to chick foraging trials with the lip of the 400-mL beaker flush with the ground. We installed 3 aluminum sheet metal strips (45 cm × 10 cm) evenly around the edge of the beaker to serve as array drift fences. We emptied each pitfall trap in the evening, closed it at night to avoid nocturnal arthropods, and reopened it the following morning. We sampled 2 trap-days for each pitfall trap. We removed arthropods from the ethylene glycol solution as soon as possible and transferred them to 120-mL cups containing ethyl alcohol to avoid degradation of arthropod diagnostic features.

We identified all arthropods collected from chick crops, sweep nets, and pitfall traps to taxonomic order using diagnostic body fragments including heads, antennae, cerci, pronota, mandibles, femora, tarsi, tibiae, wings, and body segments. We used *Borror and Delong's Introduction to the Study of Insects*, seventh edition, to classify arthropods (Triplehorn and Johnson 2005) and counted one arthropod for all body parts identified. We then calculated a total count of arthropods and mean dry weight of each order for each respective sampling method.

Vegetation Structure Sampling

To quantify effects of each vegetation management practice on vegetation structure and composition, we measured vegetative characteristics within each plot immediately following chick foraging trials. We conducted one vegetation survey at the approximate center of each plot. We recorded percentage cover of grasses, forbs, woody species, ferns, bare ground, and debris within a 0.5-m² Daubenmire frame at plot center and in each of the 4 cardinal directions 10 m from plot center (Daubenmire 1959, Greenfield et al. 2002). We measured canopy closure using a spherical forest densiometer placed on the ground at plot center and also in each of the 4 cardinal directions 10 m from plot center (Lemmon 1956). We used visual obstruction readings (VOR) determined with a Robel pole as an index to understory vegetation structure, height, and density at minimum, maximum, and average levels (Greenfield et al. 2002). We recorded VOR readings in each cardinal direction 10 m from plot center.

Statistical Methods

To quantify vegetation response relative to vegetation management practices, we performed an analysis of variance (ANOVA) using Proc MIXED (SAS Institute Inc., Cary, NC) on vegetation variables measured within each plot. We used Pearson's correlation coefficients to determine highly correlated variables, which resulted in 7 of 13 variables being retained. Retained variables included percentage of bare ground, forb, woody vegetation, grass, and vine, and minimum (VORmin) and average (VORavg) VOR. In the MIXED model, we were not interested in year or brooding-period effects; therefore, we only evaluated effects of management practices on vegetation response. We included year as a random effect and nested plot within year and vegetation management practice. We used Tukey–Kramer

Table 1. Mean (with associated SE) vegetation characteristics associated with 4 understory vegetation management practices on Jackson-Bienville Wildlife Management Area, Louisiana, USA, 2002–2005.

Variable	Management practice ^a							
	Burn		Control		Burn–herbicide		Mow	
	$\bar{x}$	SE	$\bar{x}$	SE	$\bar{x}$	SE	$\bar{x}$	SE
Grass cover (%)	14.31 ^B	0.99	9.951	1.17B	15.62	1.09B	19.82	1.29A
Forb cover (%)	11.78B	0.94	9.551	1.25B	21.12	1.11A	12.85	1.18B
Woody cover (%)	29.93B	1.73	41.98	2.71A	18.07	1.43C	24.83	1.61BC
Vine cover (%)	8.041A	0.69	5.981	0.85A	8.271	0.69A	6.711	0.65A
Debris cover (%)	17.64	1.19	29.19	2.09	21.37	1.36	34.46	1.33
Bare ground cover (%)	11.55 A	(1.11)	1.441	0.41B	11.19	0.94A	0.921	0.25B
Fern cover (%)	6.832	1.51	2.041	0.72	4.121	0.92	0.321	0.14
Canopy closure (%)	70.68	1.62	82.06	1.63	70.73	1.26	86.11	0.88
VORmin vegetation ht (m) ^c	0.471AB	0.02	0.610	0.04A	0.281	0.01BC	0.270	0.02C
VORmax vegetation ht (m) ^c	0.961	0.03	1.131	0.03	0.710	0.02	0.651	0.03
VORavg vegetation ht (m) ^c	0.610AB	0.02	0.730	0.04A	0.421	0.02BC	0.381	0.02C
Dbh	0.340	0.03	0.311	0.03	0.322	0.02	0.312	0.01
Distance to nearest tree from plot center (m)	1.991	0.21	2.092	0.20	2.011	(0.17)	2.102	(0.13)

^a Burn = prescribed fire during the growing season, Control = no treatment, Burn–herbicide = imazapyr application in combination with prescribed fire during the growing season, Mow = mowed vegetation.

^b No. within rows followed by identical letters are not different ($P \leq 0.05$); differences presented for 7 variables retained for analysis following Pearson correlation analysis.

^c VOR = visual obstruction reading.

adjustments to pairwise comparisons to evaluate differences among mean vegetation variables.

Similarly, to quantify both the overall arthropod abundance (all taxonomic orders) and abundance of individual taxonomic orders relative to vegetation management practices and brooding periods, we performed an ANOVA using the Proc MIXED procedure on sweep net and pitfall trap data. We performed one ANOVA for the overall abundance estimate for all sweep net and pitfall collections relative to vegetation management practices and brooding period. We subsequently performed individual ANOVAs for each taxonomic order to quantify abundance of arthropods by order relative to vegetation management practices and brooding period. In the MIXED model for both analyses, we included year as a random effect and nested plots where samples were collected within year, vegetation management practice, and brooding period. Subplots (3 pitfall traps and sweep nets/plot) were nested within year, vegetation management practice, brooding period, and plot. We were not interested in interactions between brooding period and vegetation management practice within these models.

To quantify the probability of chicks successfully capturing an arthropod relative to vegetation management practice and brooding period, we designed a logistic regression model using Proc GLIMMIX in SAS. The GLIMMIX procedure is a type of generalized linear mixed model that fits statistical models to data with correlations of non-constant variability and where the response is not necessarily normally distributed (Blouin and Saxton 1990). We assigned a binary ordered value of zero (unsuccessful at catching arthropod) or one (successfully caught arthropod) to each chick. In the GLIMMIX model, we used year as a random effect and nested plot within year, vegetation

management practice, and brooding period. Because brood (5 chicks/brood) was the experimental unit, we nested it within plot, year, vegetation management practice, and brooding period. Results from the GLIMMIX procedure presented odds ratio estimates, Type III tests of fixed effects, and least squares means (Saxton's macro probabilities detransformed by the GLIMMIX procedure). We used each of these results to interpret pairwise comparisons and interactions between vegetation management practice and brooding period. We performed all procedures using SAS 9.1 (SAS Institute Inc.).

To determine arthropod species diversity among vegetation management practice, we developed a Shannon–Weiner (S–W) diversity index to account for the evenness and abundance of arthropod species relative to each. We calculated the index by using the proportion of species (i), relative to the total number of species (p_i) within each management practice, multiplied by the natural logarithm of this proportion ($\ln p_i$). We summed the resulting product across species and multiplied by -1 .

RESULTS

Percentage of woody vegetation was nearly 2 times greater in control plots, whereas percentage of forbs was 2 times greater in burn–herbicide plots relative to the other practices (Table 1). Mowed plots contained the greatest percentages of grass, whereas burn and burn–herbicide plots both contained 10 times the amount of bare ground. Percentage of vine did not differ among management practices. We found that VORmin and VORavg also differed among practices, because control and burn plots contained minimum and average vegetation heights >2 times those found in burn–herbicide and mowed plots.

Arthropod abundance tended to be greater in burn–

Table 2. Mean arthropods per sample with associated standard errors from pitfall trap, sweep net, and bobwhite chick crop samples within loblolly pine stands managed with 4 understory vegetation management techniques on Jackson-Bienville Wildlife Management Area, Louisiana, USA, 2002–2005.

Yr	Treatment ^a	Pitfall samples		Sweep net samples		Chick crops	
		$\bar{x}$	SE	$\bar{x}$	SE	$\bar{x}$	SE
2002	Burn ^b						
	Control	20.55	1.25	2.66	0.45	0.19	0.08
	Burn–herbicide ^b	15.27	1.15	2.72	0.27	0.18	0.06
	Mow	21.72	1.60	5.05	1.03	0.25	0.08
2003	Burn	9.60	0.67	3.33	0.28	0.74	0.24
	Control	14.08	2.61	4.33	0.88	2.02	0.64
	Burn–herbicide	17.37	1.68	3.33	0.20	1.22	0.19
	Mow	42.33	6.80	3.61	0.27	1.34	0.38
2004	Burn	13.91	0.89	21.85	0.76	0.74	0.11
	Control	10.83	0.89	18.92	0.71	1.05	0.17
	Burn–herbicide	14.83	1.02	31.07	0.79	1.28	0.07
	Mow	9.41	0.89	16.71	0.53	1.15	0.11
2005	Burn	7.83	0.77	12.86	0.52	0.71	0.10
	Control	4.00	0.42	21.58	1.21	0.76	0.13
	Burn–herbicide	10.50	1.17	17.35	0.51	1.24	0.08
	Mow	8.20	0.59	22.03	0.88	0.65	0.14

^a Treatments: Burn = growing-season prescribed burn, Control = no treatment, Burn–herbicide = imazapyr application followed by a growing-season burn, Mow = mowed vegetation.

^b Burn treatments were conducted in 2003 and 2005; that is, 2002 burn–herbicide treatment is imazapyr-only without the growing-season burn.

herbicide and mowed plots (Table 2). Orthoptera abundance differed by management practice (with greatest abundance in burn–herbicide plots; $F_{3,78} = 3.73$, $P = 0.014$) as did Araneae abundance ($F_{3,98} = 2.62$, $P = 0.055$). We did not detect differences ($P > 0.245$) in abundance of Hemiptera, Coleoptera, Diptera, Lepidoptera, Hymenoptera, or orders in the Other category among practices (Table 2). Likewise, abundance of Hemiptera, Coleoptera, Diptera, Lepidoptera, Hymenoptera, or orders in the Other category did not differ ($P > 0.244$) by brooding period. Abundance of Orthoptera and Araneae differed by brooding period ($F_{2,78} = 4.52$, $P = 0.014$ and $F_{2,98} = 3.88$, $P = 0.024$, respectively).

Shannon–Weiner diversity indices for all vegetation management practices reflected low arthropod heterogeneity for each sampling method (Table 3). Burn–herbicide plots had the greatest arthropod diversity indices, whereas control

plots produced the least diversity relative to other management practices.

We used 1,155 imprinted bobwhite chicks to determine the probability of a chick capturing an arthropod relative to vegetation management practice and brooding periods from 2002 to 2005. During foraging trials, 479 chicks were successful and 676 were unsuccessful at catching an arthropod.

Chicks were more successful at capturing arthropods in burn–herbicide plots. Vegetation management practice and brooding period did not interact ($F_{6,94} = 1.42$, $P = 0.215$) to affect the probability of chicks capturing arthropods, but the probability of capturing an arthropod differed among management practices ($F_{3,94} = 9.98$, $P < 0.001$) and brooding periods ($F_{2,94} = 4.84$, $P = 0.010$). De-transformed probabilities and odds ratio comparisons showed that chicks successfully captured more arthropods in burn–herbicide

Table 3. Shannon–Weiner (S–W) diversity indices for arthropods collected in pitfall trap, sweep net, and bobwhite chick crop samples within loblolly pine stands managed with 4 understory vegetation management techniques at Jackson-Bienville Wildlife Management Area, Louisiana, USA, 2002–2005.

Method	Index	Treatment			
		Burn ^a	Control	Burn–herbicide	Mow
Pitfalls	Total no. of samples	72	66	126	81
	Total no. of arthropods	718	847	1,778	1,449
	No. arthropods/sample	9.9	12.8	14.11	17.9
	Diversity (H') ^b index	1.223	1.047	1.252	1.165
Sweep nets	Total no. of samples	72	69	126	87
	Total no. of arthropods	972	813	2,121	1,207
	No. arthropods/sample	13.5	11.8	16.8	13.9
	Diversity (H') index	1.691	1.632	1.741	1.686
Chick crops	Total no. of chicks	247	216	405	287
	Total no. of broods	50	44	83	58
	Total no. of arthropods	179	204	444	237
	Diversity (H') index	0.968	0.831	1.365	1.135

^a Burn treatments were not conducted prior to 2003, so indices and totals for this treatment are only for 2003–2005.

^b We calculated the S–W diversity index as $H' = \sum [p_i \times \ln(p_i)]$.

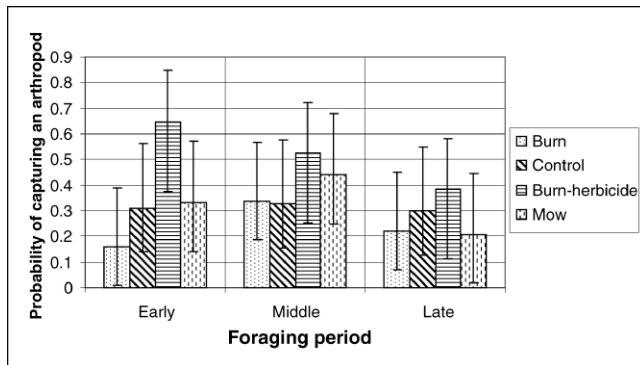


Figure 1. Imprinted bobwhite chick foraging probabilities with associated confidence intervals within loblolly pine stands managed with 4 understory vegetation management techniques in early (Jun), middle (Jul), and late (Aug) brooding periods for northern bobwhites on Jackson-Bienville Wildlife Management Area, Louisiana, USA, 2002–2005.

plots (Fig. 1). For instance, the probability of capturing an arthropod in burn–herbicide plots was 137% greater than in control plots (Table 4). Probability of capturing an arthropod did not differ between early and late (Aug) brooding periods but was 82% greater during the middle brooding period (Jul) than during the late (Aug) period (Table 5), coinciding with the peak in brooding activity for bobwhites in north Louisiana.

DISCUSSION

We observed that management of understory vegetation using a combination of prescribed fire and imazapyr applications improved brooding habitat for bobwhites, primarily through maintenance of herbaceous and forbaceous plant communities, which positively influenced invertebrate response and the probability that bobwhite chicks successfully captured arthropods. High-quality brood habitat for bobwhites typically contains great plant species richness, a prevalence of forbaceous plants, and sufficient bare ground to allow chicks to forage freely and find invertebrates (Burger 2001, Palmer et al. 2001). Imazapyr application with prescribed fire improved herbaceous and forbaceous plant communities, relative to other management practices, and increased bare ground. Although using prescribed fire alone typically reduces hardwood stem densities and improves early successional ground cover, it

Table 4. Pairwise comparisons with associated confidence intervals using odds ratios to evaluate bobwhite chick foraging efficiency among loblolly pine stands managed with 4 understory vegetation management techniques on Jackson-Bienville Wildlife Management Area, Louisiana, USA, 2002–2005.

Treatment comparison	Estimate	95% CI		Difference
		Lower	Upper	
Mow vs. burn–herbicide	0.435	0.281	0.675	–57%
Mow vs. control	1.032	0.637	1.673	+3%
Mow vs. burn	1.552	0.943	2.555	+55%
Burn–herbicide vs. control	2.372	1.470	3.828	+137%
Burn–herbicide vs. burn	3.567	2.163	5.884	+256%
Control vs. burn	1.504	0.876	2.582	+50%

Table 5. Pairwise comparisons with associated confidence intervals using odds ratios of bobwhite chick foraging efficiency among 3 periods of the brood-rearing season for northern bobwhites on Jackson-Bienville Wildlife Management Area, Louisiana, USA, 2002–2005.

Period comparison ^a	Estimate	95% CI		Difference
		Lower	Upper	
Middle vs. late	1.821	1.242	2.670	+82%
Middle vs. early	1.289	0.826	2.012	+29%
Late vs. early	0.708	0.444	1.129	–30%

^a Early = Jun, middle = Jul, late = Aug.

often takes repeated years of burning to restore and maintain this ground cover (Cain et al. 1998). However, using prescribed fire in combination with imazapyr applications typically reduces woody vegetation while stimulating production of herbaceous plants, forbs, and legumes (Watkins et al. 1989, Jones and Chamberlain 2004, Welch et al. 2004). Arthropods comprise >80% of bobwhite chick diets during the first 2 weeks of life and provide proteins, amino acids, water, and energy needed for survival and growth (Nestler 1940, Hurst 1972, Jackson et al. 1987). Manipulating composition and structure of vegetation can directly influence diversity, abundance, and availability of arthropods to young bobwhite chicks, and arthropod populations within forested environments typically increase with increases in herbaceous vegetation (Healy 1978, Southwood et al. 1979). Imazapyr application with prescribed fire tended to increase abundance and diversity of arthropods, presumably through improvements in early successional ground cover. Improvements to vegetation structure and composition, coupled with increased abundance and diversity of arthropods, suggest substantial improvements to brood habitat quality in stands managed with imazapyr and prescribed fire. However, arthropod abundance in our study (no. arthropods/sample = 2.661–42.331) was substantially less compared to previous studies in other landscape types (no. arthropods/sample = 10.2–238.3; Hurst 1972, Jackson et al. 1987, Welch 2000, Palmer et al. 2001).

Although they likely reflect the magnitude of arthropod response relative to the treatments we evaluated, our estimates of arthropod abundance were not accurate assessments of arthropod availability to wild bobwhite broods. During foraging bouts, wild chicks are subjected to a variety of factors, such as inclement weather, habitat composition and structure, predation, disease, and arthropod abundance and availability (Stoddard 1931, Rosene 1969, Burger et al. 1993). Both Palmer et al. (2001) and Smith and Burger (2005) determined that the use of human-imprinted pen-strain bobwhite chicks was a reasonable biological assay for more accurate indices regarding foraging quality and arthropod availability as they pertain to wild bobwhite chicks. We inherited assumptions of these studies, namely that chick foraging behaviors were innate and that they provided valuable information as to availability and selection of arthropods by wild chicks. Applying imazapyr with prescribed fire increased the probability of chicks capturing

arthropods, which, coupled with improvements in vegetation structure and arthropod abundance and diversity, suggests that managing understory vegetation in pine forests similar to those we studied with combinations of imazapyr and fire improves bobwhite brooding habitat.

Notably, only 41% (479 of 1,155) of imprinted chicks in our study were successful at consuming arthropods during a 30-minute foraging trial. Furthermore, chicks in our study did not consume as many arthropods (mean no. arthropods/chick = 0.183–2.023) as chicks in other studies (e.g., mean arthropods/chick = 8.4–29.5; Palmer et al. 2001, Smith and Burger 2005), although we recognize that our estimates of numbers of arthropods consumed are likely underestimates. Palmer et al. (2001) examined crops and gizzards of imprinted chicks, and noted that 47.1% of arthropods recovered were from gizzards. Similarly, Smith and Burger (2005) noted that 33% of arthropod biomass passed to the gizzard. Therefore, we would expect our samples to have fewer arthropods per chick because we did not formally include gizzards in our analysis. However, during the initial year of our study we examined a sample of gizzards (approx. 10%) to ensure that samples in which we detected no arthropod fragments in the crop were not a result of passage of arthropod remains to the gizzard. We did not detect remains in the gizzards either, so we felt comfortable with our analysis because it was binary in response (i.e., a chick either was successful or not). Regardless, if a bias exists relative to arthropods passing from the crop to the gizzard, we suspect it is distributed equally across vegetation management practices and should not affect our interpretation of how understory vegetation management affects quality of bobwhite brooding habitats.

Although managing understory vegetation with prescribed fire and imazapyr improved vegetation structure and arthropod abundance, poor foraging success of chicks was concerning. We hypothesize that poor foraging success was most likely related directly to vegetation structure not conducive to optimizing foraging success. Despite improvements to vegetation following all of the vegetation management practices, plots still did not contain adequate amounts of bare ground. Bare ground is an important component of bobwhite brood habitat, and the proportion of bare ground in the understory is optimal at 25–75% for nesting and brood-rearing habitats (Rice et al. 1993, Taylor and Burger 2000). Furthermore, debris (primarily needlecast) was a prominent component of understory structure (17–34% of understory composition) and likely compromised the ability of chicks to successfully capture arthropods. We offer that creating and maintaining optimal brood habitats in pine forests similar to those we studied is more difficult relative to other habitats. Previous studies in our region have suggested that additional disturbances (e.g., mechanical) or more intensive fire regimes may be required to maintain bare ground at suitable levels in managed pine forests (Jones and Chamberlain 2004).

Brood ecology is the least understood component of bobwhite population ecology (Burger 2001). Optimum

growth and survival of chicks requires a high-protein diet, and protein content may affect immune response of chicks (Nestler et al. 1942, Lochmiller et al. 1993). Reduced abundance of arthropods may negatively influence growth and survival of gray partridge (*Perdix perdix*), which in turn can affect population growth rates (Hill 1985, Potts 1986). Within managed pine forests similar to those we studied, brood habitat quality may be similarly affecting bobwhite population growth. During our study 1,666 ha (12% of total study area) of mature (>35 yr) pine forest was converted into stands managed for RCW, which included the use of prescribed fire on a 2-year rotation or the use of fire in combination with imazapyr (as detailed herein). Despite these habitat improvements at the forest stand level, abundance of breeding bobwhites (estimated using surveys of calling males) declined 29% during this same period (Kitts 2004, Burke 2006).

MANAGEMENT IMPLICATIONS

The Northern Bobwhite Conservation Initiative (NBCI) promotes the altering of silvicultural methods on approximately 50 million acres of forest lands, primarily southern pine forests, to encourage favorable grass and forb communities (Dimmick et al. 2002). Our findings suggest that the application of imazapyr in combination with a 2-year prescribed fire rotation, which in our study was directed at renovating habitats for RCWs, could play an important role in landscape-level improvement of southern pine forests as proposed by the NBCI. However, clearly more action is needed. Managers should recognize that pine forests similar to those we studied likely fail to provide quality brooding habitat for bobwhites, and more intensive management of understory vegetation than we evaluated is necessary. Future research should evaluate the intensity of vegetation management necessary to achieve high-quality brood habitat in loblolly pine-dominated landscapes.

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